PhonoProbe: Do Speech Models Hear Through Neutralization?

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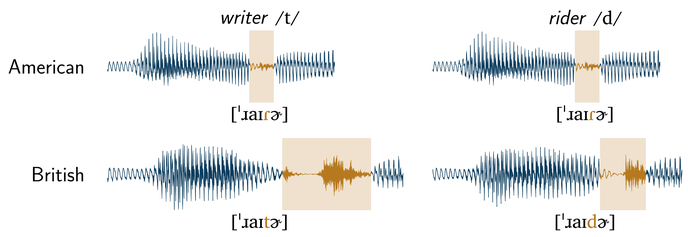
Demo video: devans2718.github.io/phonoprobe-video

Code: github.com/devans2718/capstone

Problem statement

How do we compare what self-supervised speech models represent about speech? PhonoProbe tests this on one hard case: neutralization, where the sound on the surface erases a contrast the language keeps underneath.

In American English, /t/ and /d/ between vowels often collapse into the same quick flap, [ɾ], so *writer* and *rider* can sound alike.



Synthesized (Kokoro-82M), flap forced in the American pair; British keeps the two apart.

Do self-supervised speech encoders keep the underlying /t/ or /d/, or only the sound on the surface? The same question applies well beyond English:

- Final devoicing: German *Rad* 'wheel' and *Rat* 'advice' are both [ʁaːt]; Russian код 'code' and кот 'cat' are both [kot]
- Inflection: the accusative of кот is кота, but the accusative of код is код, the same as its nominative

Can models recover phonological structure that neutralization hides? If so, they could support comparative and historical linguistics, or help draft descriptive grammars of under-documented languages.

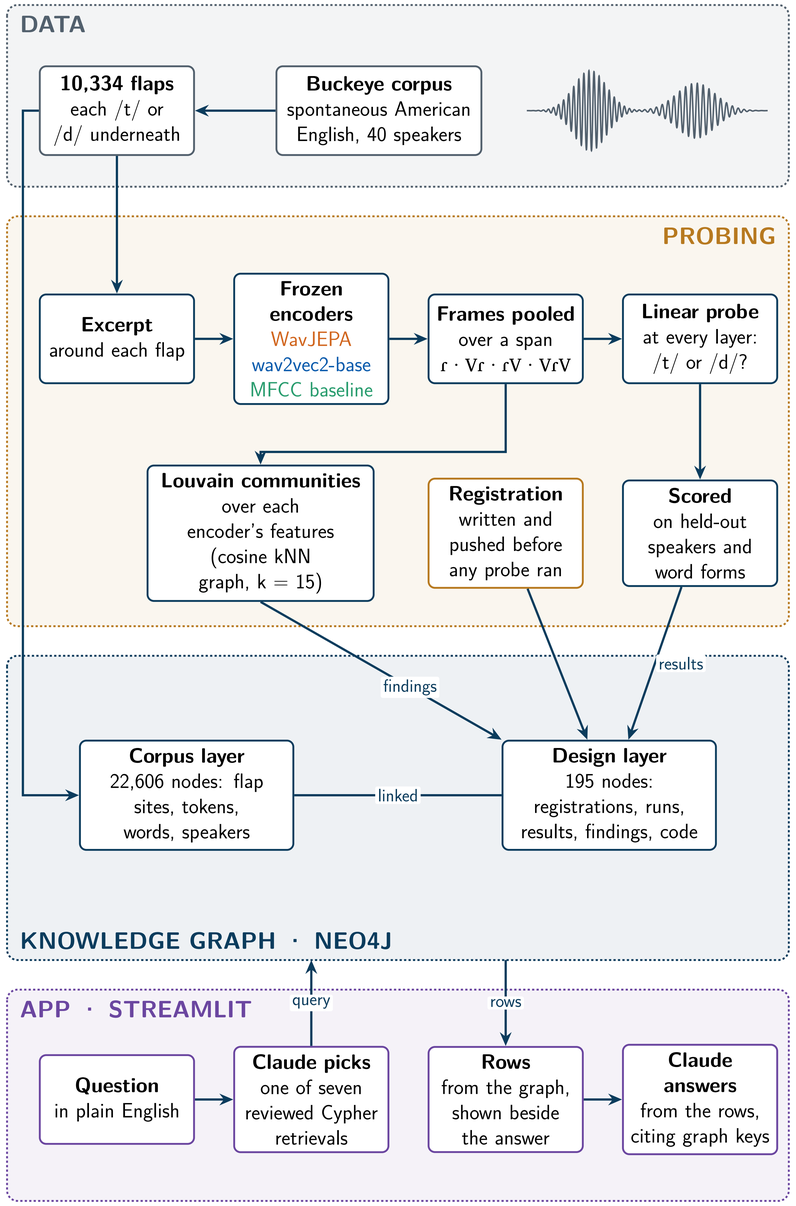
People and AI

- The knowledge graph tied each run to the right code commits; one run's two commits turned out to be two pipeline stages
- AI agents wrote code from briefs I approved; I set the questions, made the calls, and reviewed each result before it was merged or recorded

Where this goes

- HuBERT and WavLM on the same protocol
- Beyond flapping: final devoicing and vowel mergers, then inflection and syntax
- Next study: a lighter knowledge graph from the design stage, not built after the fact

How PhonoProbe works



When the first probe failed

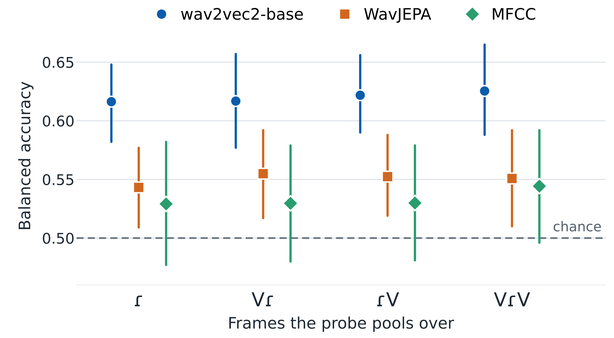
The first pre-registered probe, on WavJEPA, met 1 of its 4 criteria, close to chance at every layer. One model near chance can't say whether the model or the method failed, so comparisons came first, some already planned and some new: a second encoder, more baselines, and a span grid.

Try it: ask the knowledge graph

- Answers come only from the returned rows, shown beside them
- A communities retrieval answered 6 of 6 community questions (all 6 refused without it), with 0 of 18 wrong picks elsewhere

Try asking: *"Why did the pilot fail?"* The answer lists the failed criteria, the candidate causes with their status, and what each rests on.

Probe accuracy by encoder and span



Best layer for each encoder and span; bars are 95% intervals, resampled by word type. In this setup wav2vec2-base (layer 9, about 0.62) leads MFCC by 0.08–0.09; WavJEPA (layers 3–6, about 0.55) beats no baseline. V before the flap is the preceding sonorant stretch; V after it is the next segment. Each encoder heard the same excerpt for every span, and widening the pooled span changed no representation by more than 0.015; cropping the input to the span, not yet run, could still help the encoders. All control tasks $p > 0.05$.

What the encoders group by

	Phonetic environment	Speaker	Word	/t/ vs /d/
wav2vec2-base L9	0.089	0.030	**0.096**	0.007
WavJEPA L3	0.043	**0.234**	0.045	0.002
MFCC	**0.160**	0.126	0.142	0.007

Adjusted mutual information between Louvain communities over each representation's flap features and each grouping (phonetic environment: the vowel and following segment); largest in each row in bold. Exploratory, not registered; no probe trained; not the knowledge graph's communities.

What it means: the encoders' communities lean toward word (wav2vec2-base) or speaker (WavJEPA); only MFCC's lean most toward phonetic environment. Next: can training or fine-tuning make a model organize around phonetic environment instead?

Why beat chance at all? Possibly a shortcut, not a leak: a flap's word fixes its label, and the encoders hear the neighboring words. Not yet tested.